

Spivak Chapter 11 Problems

March 5, 2026

1 Problem 1

For each of the following functions, find the maximum and minimum values on the indicated intervals.

1. $f(x) = x^3 - x^2 - 8x + 1$ on $[-2, 2]$.
 $f'(x) = 3x^2 - 2x - 8$, $f'(x) = 0$ at $x = 2, -\frac{4}{3}$. $f(x)$ has a local minimum at 2 and a local maximum at $-\frac{4}{3}$.
2. $f(x) = x^5 + x + 1$ on $[-1, 1]$.
 $f'(x) = 5x^4 + 1$, $f'(x) = 0$ has no real solutions. $f(x)$ has an absolute minimum at -1 and an absolute maximum at 1 .
3. $f(x) = 3x^4 - 8x^3 + 6x^2$ on $[-\frac{1}{2}, \frac{1}{2}]$.
 $f'(x) = 12x^3 - 24x^2 + 12x$, $f'(x) = 0$ at $x = 0, 1$. $f(x)$ has an absolute minimum at 0 and an absolute maximum at $-\frac{1}{2}$.

2 Problem 6

1. Let (x_0, y_0) be a point of the plane, and let L be the graph of the function $f(x) = mx + b$. Find the point $\bar{x}$ such that the distance from (x_0, y_0) to $(\bar{x}, f(\bar{x}))$ is smallest.
 $D^2 = (\bar{x} - x_0)^2 + (m\bar{x} + b - y_0)^2$,
 $\frac{d}{d\bar{x}}D^2 = 2(\bar{x} - x_0) + 2m(m\bar{x} + b - y_0)$,
 $\bar{x} = m(m\bar{x} + b - y_0) + x_0$,
So the closest point is $(\bar{x}, m\bar{x} + b)$.

3 Problem 8

A straight line is drawn from the point $(0, a)$ to the horizontal axis, and then back to $(1, b)$. Prove that the total length is shortest when the angles α and β are equal.

$$D = \sqrt{x^2 + a^2} + \sqrt{(1-x)^2 + b^2},$$

$$D' = \frac{x}{\sqrt{x^2 + a^2}} - \frac{1-x}{\sqrt{(1-x)^2 + b^2}},$$

Then

$$\cos(\alpha) = \frac{x}{\sqrt{x^2 + a^2}}$$

and

$$\cos(\beta) = \frac{1-x}{\sqrt{(1-x)^2 + b^2}}$$

Setting $D' = 0$, we have $\cos(\alpha) = \cos(\beta)$.

4 Problem 9

Prove that of all rectangles with given perimeter, the square has the greatest area.

$$\begin{aligned} p &= 2l + 2w, \\ a &= lw, \\ l &= \frac{p-2w}{2}, w = \frac{p-2l}{2}, \\ a &= \frac{l(p-2l)}{2} \\ a' &= \frac{p-4l}{2} \\ \frac{p-4l}{2} &= 0 \\ p &= 4l, \text{ a square.} \end{aligned}$$

5 Problem 10

Find, among all right circular cylinders of fixed volume V , the one with the smallest surface area.

$$\begin{aligned} V &= \pi r^2 h, S = 2\pi r(r + h), \\ h &= \frac{V}{\pi r^2}, \\ S &= 2\pi r^2 + \frac{2V}{r}, \\ S' &= 4\pi r - \frac{2V}{r^2}, \\ 4\pi r - \frac{2V}{r^2} &= 0, \\ r &= \frac{\sqrt[3]{4\pi^2 V}}{2\pi}, h = 2r. \end{aligned}$$

6 Problem 25

1. Prove that if $f'(x) \geq M$ for all x in $[a, b]$, then $f(b) \geq f(a) + M(b - a)$.

Applying the Mean Value Theorem, there is an $f'(x)$ such that

$$f'(x) = \frac{f(b) - f(a)}{b - a} \geq M,$$

so

$$f(b) - f(a) \geq M(b - a),$$

$$f(b) \geq f(a) + M(b - a).$$

2. Prove that if $f'(x) \leq M$ for all x in $[a, b]$, then $f(b) \leq f(a) + M(b - a)$.

Similarly, there is an $f'(x)$ such that

$$f'(x) = \frac{f(b) - f(a)}{b - a} \leq M,$$

so

$$f(b) - f(a) \leq M(b - a),$$

$$f(b) \leq f(a) + M(b - a).$$